

BAHARIA, Egypt

WMO: 624200

Lat: 28.333N

Lon: 28.867E

Elev: 130

StdP: 99.77

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	6.1	-7.2	2.1	19.8	-5.2	2.5	18.7	7.1	16.3	6.4	16.8	0.5	200	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	42.0	20.4	40.6	20.3	39.3	20.3	22.9	33.5	22.4	33.5	22.0	33.4	3.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.5	25.9	19.6	14.6	25.9	18.7	13.8	25.9	68.4	33.1	66.4	33.3	64.6	33.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	3.2	44.4	1.6	1.9	2.1	45.8	1.2	46.9	0.3	48.0	-0.8	49.3		
(5)			0.4	24.1	1.4	0.7	-0.6	24.6	-1.4	25.1	-2.2	25.5	-3.2	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	13.8	15.5	19.1	23.0	26.8	29.7	30.9	31.2	28.8	24.8	19.7	15.2		
	DBStd	6.72	2.20	3.12	3.59	3.86	3.23	2.55	1.88	1.66	2.28	2.70	2.63	2.35		
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	380	142	89	31	3	0	0	0	0	0	0	14	101		
	CDD10.0	4836	118	156	282	391	522	590	647	657	564	457	289	163		
	CDD18.3	2174	1	11	55	144	264	340	389	398	314	199	53	5		
	CDH23.3	29293	30	149	704	1797	3573	4969	5838	5866	3982	1903	428	54		
(13)	CDH26.7	17201	4	48	292	913	2086	3142	3737	3692	2303	862	112	11		
(14)	Wind	WSAvg	2.0	1.7	1.9	2.2	2.4	2.6	2.3	2.0	2.2	1.9	1.5	1.6		
(15) Precipitation	PrecAvg	3	1	1	1	1	0	0	0	0	0	0	0	0		
	PrecMax	14	4	8	7	12	7	0	0	0	1	2	5	2		
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	4	1	2	2	2	1	0	0	0	0	0	1	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	31.6	36.8	40.3	43.7	44.4	43.7	42.9	41.1	38.1	32.9	28.1		
		MCWB	13.4	15.4	17.0	18.3	19.3	19.7	20.8	21.1	20.7	20.3	18.1	16.1		
	2%	DB	23.5	27.6	33.0	37.3	40.2	41.7	41.2	41.0	39.3	35.3	29.4	24.5		
		MCWB	12.6	13.4	15.6	17.5	18.9	20.1	20.7	20.8	20.5	19.5	17.2	14.1		
	5%	DB	21.4	24.5	29.8	34.7	37.7	39.7	39.8	39.6	37.6	33.2	27.4	22.3		
		MCWB	11.9	12.6	14.7	16.5	18.3	19.8	20.6	21.0	20.4	19.0	16.5	13.6		
	10%	DB	19.7	22.5	27.0	31.9	35.6	37.8	38.3	38.2	35.8	31.2	25.6	20.9		
		MCWB	11.5	12.1	13.8	16.0	17.9	19.7	20.5	20.9	20.2	18.7	15.9	13.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	16.2	18.6	19.8	20.7	22.3	23.2	24.0	23.1	22.2	19.7	16.7		
		MCDB	23.8	28.9	31.7	36.0	37.8	36.7	34.9	32.6	32.2	33.4	28.3	25.3		
	2%	WB	13.6	14.4	16.5	18.4	19.8	21.6	22.6	23.2	22.2	21.0	18.4	15.2		
		MCDB	21.0	24.6	30.3	34.6	36.0	36.5	34.6	33.0	32.0	30.7	26.8	22.3		
	5%	WB	12.7	13.4	15.3	17.3	19.2	21.0	22.1	22.6	21.7	20.2	17.4	14.3		
		MCDB	20.0	22.6	27.5	32.5	34.7	35.1	34.2	33.1	31.8	29.3	25.3	21.2		
	10%	WB	12.0	12.6	14.4	16.2	18.6	20.4	21.6	22.1	21.2	19.5	16.6	13.4		
		MCDB	19.0	21.4	25.4	30.1	33.3	34.2	33.8	33.1	31.7	28.3	24.1	20.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	13.2	14.5	15.0	15.6	15.7	15.1	14.4	14.0	13.1	12.7	12.3		
		MCDBR	15.2	16.3	17.7	18.4	18.4	17.9	16.8	16.1	16.3	15.5	14.5	14.0		
	5% WB	MCWBR	7.8	7.2	6.9	6.4	5.7	4.7	4.7	4.5	4.6	5.1	6.0	6.9		
		MCWBR	13.2	14.2	16.0	16.3	15.8	15.6	14.8	14.1	14.0	13.3	13.1	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.393	0.435	0.481	0.556	0.562	0.488	0.466	0.460	0.467	0.454	0.404	0.389		
	taud		2.036	1.907	1.789	1.649	1.664	1.849	1.922	1.947	1.920	1.940	2.046	2.061		
	Ebn at Noon		824	820	808	762	757	811	828	831	810	790	805	806		
	Edh at Noon		145	179	214	255	251	208	193	186	186	171	142	135		
(44) All-Sky Solar Radiation	RadAvg		4.02	4.92	6.15	7.04	7.69	8.15	7.97	7.45	6.60	5.38	4.30	3.74		
	RadStd		0.09	0.21	0.16	0.17	0.17	0.11	0.06	0.08	0.12	0.08	0.10	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.57	+0.75	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page